

Study Notes and Quick Reference

Point Processing: Histogram Equalisation and CLAHE

Core focus: CDF derivation, worked example, CLAHE as adaptive improvement, and failure cases.

1. Big Picture: What Problem Are We Solving?

Many useful images are technically correct but visually hard to read. A face camera may capture a person in uneven lighting, a spine X-ray may have low contrast, and a dark image may hide edges that are important for later recognition or segmentation. Histogram equalisation and CLAHE are point processing techniques used to make intensity differences more visible. They do not create new scene information; they remap existing pixel values so that useful details become easier to see or easier for an algorithm to use.

A helpful analogy is a classroom with students whispering, speaking normally, and shouting. If most people whisper, it is difficult to tell voices apart. Histogram equalisation works like spreading the volume levels across a wider range so quiet voices become more distinct. CLAHE is the more careful version: instead of adjusting the whole classroom with one volume knob, it adjusts small groups locally and stops any group from becoming too loud.

Point processing means each output pixel is produced from the intensity value of an input pixel, usually using a transformation function. For a basic pixel transform, $g(x,y) = T[f(x,y)]$. The uploaded point-processing paper defines this idea as a transformation that maps $f(x,y)$ to $g(x,y)$ at the same coordinate, affecting only that pixel location. Histogram equalisation is slightly more global than simple brightness adjustment because the mapping function T is built from the image histogram, but once the mapping is created, each pixel is still remapped by a lookup table.

2. Point Processing Foundations

A digital grayscale image can be treated as a two-dimensional function $f(x,y)$, where x and y are spatial coordinates and f is the intensity value. For an 8-bit image, each pixel intensity is usually an integer from 0 to 255. Low values are dark, high values are bright. In colour images, the same idea can apply to each RGB channel, but for histogram equalisation it is often safer to work on a luminance channel rather than equalising RGB channels separately.

Common point operations include negative transformation, thresholding, brightness change, contrast stretching, histogram equalisation, and local contrast enhancement. The point-processing source explains that such filters are attractive because they are simple and fast, especially when many pixels can be processed in parallel on a GPU. For example, thresholding maps a pixel to black or white based on whether it is below or above a chosen threshold.

- Brightness adjustment shifts or scales intensities. Example: $s = ar + b$.
- Contrast stretching spreads a chosen intensity interval over a wider output interval.
- Thresholding converts grayscale into two levels, usually 0 and 255.
- Histogram equalisation builds a mapping from the cumulative distribution of the whole image.
- CLAHE builds many local mappings and limits their strength to avoid over-amplification.

3. Histograms: The Starting Point

A histogram counts how many pixels have each intensity. In an 8-bit grayscale image there are usually 256 histogram bins, one for each intensity level. A dark image often has most counts on the left side of the histogram. A bright image often has most counts on the right. A low-contrast image often has counts squeezed into a narrow region. The aim of histogram equalisation is to spread the intensity distribution so that the output uses more of the available range.

Let r_k be an input intensity level, where $k = 0, 1, \dots, L-1$ and L is the number of possible levels. For an 8-bit image, $L = 256$. Let n_k be the number of pixels with intensity r_k . Let N be the total number of pixels. The normalised histogram is $p(r_k) = n_k / N$. This is a probability distribution: it tells us the probability that a randomly chosen pixel has intensity r_k .

The histogram alone tells us how many pixels fall at each level, but histogram equalisation needs the cumulative distribution function, or CDF. The CDF tells us how much of the image lies at or below each intensity. It is the running total of histogram probabilities.

4. CDF Derivation for Histogram Equalisation

The core idea is to use the CDF as the transformation function. For discrete grayscale levels, the cumulative distribution up to level k is:

$$\text{CDF}(r_k) = \sum_{j=0}^k p(r_j)$$

Because $p(r_j)$ values sum to 1, the CDF begins near 0 and ends at 1. To convert this cumulative probability into an output grayscale value, multiply it by the maximum output intensity $L-1$ and round or floor the result:

$$s_k = \text{round}((L - 1) * \text{CDF}(r_k))$$

This equation means that intensity levels with many pixels cause the CDF to rise quickly, producing a larger jump in output intensity. Intensity levels with few pixels cause small changes. In theory, this makes the output histogram more spread out. In practice, because images are discrete and histograms are not perfectly continuous, the output histogram is not always perfectly flat, but the visible contrast is often improved.

The derivation can be understood as a probability remapping. Suppose r is a random variable representing input intensity. The transform $s = \text{CDF}(r)$ maps values into the range 0 to 1. In continuous probability theory, this produces a variable with an approximately uniform distribution. For digital images, we scale that result back to 0 to 255. This is why the CDF is not just a convenient trick; it is mathematically connected to flattening the intensity distribution.

5. Worked Example: Small 4-Level Image

Assume a tiny grayscale system with only four intensity levels: 0, 1, 2, and 3. This is easier to calculate by hand than a full 256-level image. Suppose an image has 10 pixels with the following histogram:

Intensity r_k	0	1	2	3
Count n_k	1	2	4	3
Probability $p(r_k)$	0.10	0.20	0.40	0.30
CDF	0.10	0.30	0.70	1.00
Mapped $s_k = \text{round}(3 \cdot \text{CDF})$	0	1	2	3

In this very small example, the final mapped levels happen to stay as 0, 1, 2, and 3 because there are only four output levels. The important step is the running total: 0.10, then 0.30, then 0.70, then 1.00. The CDF is monotonic, meaning it never decreases. That is essential because the mapping should preserve order: a brighter input should not become darker than a lower input.

Now consider a more compressed example where all pixels are in the middle levels. Suppose the counts are [0, 5, 5, 0]. The probabilities are [0, 0.5, 0.5, 0], and the CDF is [0, 0.5, 1.0, 1.0]. Mapping with $\text{round}(3 \cdot \text{CDF})$ gives [0, 2, 3, 3]. Therefore, input level 1 becomes 2 and input level 2 becomes 3. The image becomes brighter and more separated, because the two used levels are pushed apart.

For a real 8-bit image, the same process is used with 256 bins. The common implementation steps are: compute the histogram, normalise it, compute the CDF, create a lookup table, then replace every pixel by its mapped value.

6. Algorithm for Global Histogram Equalisation

A practical algorithm can be written as follows:

Step	Action
1	Convert image to grayscale or choose a luminance channel.
2	Count how many pixels fall into each intensity bin.
3	Divide each count by total pixels to get probabilities.
4	Compute the cumulative sum to obtain the CDF.
5	Scale the CDF to the output range and use it as a lookup table.

In Python/OpenCV, this is often hidden inside a function such as `equalizeHist` for grayscale images. Conceptually, however, the function is doing exactly the CDF process above. For colour images, a common safer approach is to convert RGB/BGR to a colour space such as HSV, YCrCb, or LAB, equalise the brightness/luminance channel, and then convert back. Equalising RGB channels independently can shift colours because each channel gets a different mapping.

Global histogram equalisation is useful when the whole image suffers from similar contrast problems. It is less suitable when one part is too dark and another part is already bright, because one global mapping may help one area while damaging another.

7. CLAHE as an Adaptive Improvement

CLAHE stands for Contrast-Limited Adaptive Histogram Equalization. It improves on global histogram equalisation in two main ways. First, it is adaptive: the image is divided into small tiles, and a histogram equalisation mapping is computed for each tile. This helps when illumination varies across the image. Second, it is contrast-limited: it clips tall histogram bins before computing the CDF, preventing small intensity differences and noise from being amplified too aggressively.

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The CLAHE review paper explains that CLAHE operates on small regions called tiles rather than the whole image, limits contrast mainly in homogeneous areas to avoid amplifying noise, and combines neighbouring tiles to reduce visible boundaries. The Reza paper gives the deeper mathematical and hardware view: each region has a histogram, the histogram is clipped using a clip limit, excess counts are redistributed, CDF mappings are calculated, and pixel values are obtained by combining nearby regional mappings through interpolation.

The main CLAHE pipeline is:

Stage	Meaning
1. Tile division	Split the image into a grid such as 8 x 8 tiles.
2. Local histograms	Compute one histogram for each tile.
3. Clipping	Limit bins that exceed the clip limit.
4. Redistribution	Spread clipped excess counts across other bins.
5. Local CDF mapping	Compute a CDF-based mapping for each tile.
6. Interpolation	Blend neighbouring tile mappings to avoid block artifacts.

The tile grid size controls how local the enhancement is. Smaller tiles can reveal small details but may also amplify noise and create a more artificial look. Larger tiles behave more like global histogram equalisation. The clip limit controls how strong the local contrast boost can become. A higher clip limit gives stronger enhancement but increases the risk of noise amplification and over-enhancement.

8. CLAHE Interpolation and Why It Matters

If each tile were equalised independently with no blending, boundaries between tiles could become visible. One tile might map a gray value to 80 while the neighbouring tile maps the same gray value to 120. At the tile border, this produces a sudden jump that appears as a block artifact. CLAHE avoids this by interpolating between mappings from nearby tiles.

For an inner pixel, the output is usually calculated from the four closest tile mappings. The closer a pixel is to a tile centre, the more influence that tile has. Reza describes this as a bilinear interpolation of nearby contextual regions. Border and corner pixels use fewer neighbouring regions because they do not have four surrounding tiles. This is one reason CLAHE is more complex than global histogram equalisation: it is not only computing many histograms, but also blending their mappings carefully.

In simple terms, interpolation is like asking nearby teachers to vote on a student grade, but giving more weight to the teachers closest to that student. It keeps the local adaptation while preventing sudden changes from tile to tile.

9. Failure Cases and Limitations

Histogram equalisation and CLAHE are useful, but they are not magic. They can make images look clearer, but they can also make them worse if the assumptions do not match the image.

Global histogram equalisation can fail when the image has uneven lighting. If a face is half in shadow and half under bright light, one global CDF may not fix both regions. It can also shift brightness in an unwanted way, making the whole image look washed out or unnatural. In medical images, it may exaggerate background noise or distract from clinically important structures. The CLAHE face-recognition review gives an example where global histogram equalisation improves the background but loses information in the main statue/face region because of over-brightness.

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CLAHE reduces some of those problems but introduces its own risks. If the tile size is too small or the clip limit is too high, noise can become more visible. In smooth areas, random sensor noise may be treated as local detail and enhanced. If interpolation is weak or parameters are badly chosen, tile boundary artifacts may appear. CLAHE can also over-enhance textures, making an image look crunchy or unnatural. The 2025 spine X-ray paper is a good reminder that enhancement does not always improve downstream AI performance: CLAHE and other methods significantly helped U-Net in that study, but did not noticeably improve Mask R-CNN.

Method	Common failure	Reason
Global HE	Brightness shift	One mapping is forced onto the whole image.
Global HE	Noise amplification	Rare intensity differences may be stretched too far.
Global HE	Poor uneven-lighting result	Different regions need different mappings.
CLAHE	Noise amplification	Small tiles may enhance local noise as detail.
CLAHE	Block or halo artifacts	Tile mappings may differ too strongly or be poorly interpolated.

10. Applications and When to Use Each Method

Histogram equalisation is suitable for quick global contrast improvement when lighting is fairly consistent. It is simple, fast, and easy to explain. It can be useful as a preprocessing step, a visualisation tool, or a baseline contrast enhancement method.

CLAHE is better when the image has local contrast problems: faces under changing illumination, X-ray images with uneven brightness, microscopic images, low-light surveillance frames, or scenes where important details occupy only part of the image. The CLAHE face-recognition paper uses it to handle lighting variation before LBPH face recognition. The real-time CLAHE paper focuses on medical imaging and X-ray sequences where image quality must be improved without waiting for slow offline processing. The 2025 spine X-ray segmentation paper evaluates CLAHE, HE, and anisotropic diffusion as preprocessing before U-Net and Mask R-CNN segmentation.

A good practical rule is: use global HE when the whole image has the same problem; use CLAHE when different regions have different contrast problems; avoid both or tune carefully when the image is already noisy or when downstream AI models have already learned robust features.

11. Quick Reference Cheat Sheet

Concept	Quick meaning
Point processing	Output pixel is computed from input pixel intensity using a mapping.
Histogram	Counts of pixels at each intensity level.
Normalised histogram	Histogram divided by total number of pixels; estimates probability.
CDF	Running sum of histogram probabilities.
HE mapping	$s_k = \text{round}((L-1) * \text{CDF}(r_k))$.
Lookup table	Precomputed mapping from old intensities to new intensities.
Global HE	One histogram and one CDF for the whole image.
AHE	Local histogram equalisation for neighbourhoods or regions.
CLAHE	AHE with clipping to limit contrast amplification.
Tile grid size	Number/size of local regions used by CLAHE.
Clip limit	Maximum allowed histogram bin height before redistribution.

Interpolation	Blending neighbouring tile mappings to avoid block boundaries.
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Parameter guidance: start with a moderate tile grid such as 8 x 8 and a low to moderate clip limit. Increase clip limit only if the result is too flat. Increase tile size if the result is noisy or blocky. Decrease tile size if important small local details are still hidden. Always compare the enhanced image with the original and with the needs of the later task.

Exam-style explanation: Histogram equalisation improves contrast by using the CDF of the image histogram as a transformation function. The CDF maps frequently occurring intensity ranges across a wider output range, making details more visible. CLAHE improves global histogram equalisation by applying the process locally to image tiles and clipping histograms before computing the CDF, reducing the chance of excessive noise amplification.

12. Deeper Technical Notes for Implementation

A normal implementation stores the equalisation mapping as a lookup table. This matters because the expensive part is not changing each pixel; it is building the histogram and CDF. Once the table is ready, every pixel replacement is fast: $\text{output}[y,x] = \text{LUT}[\text{input}[y,x]]$. For an 8-bit image the table has only 256 entries, so it is tiny. For 10-bit, 12-bit, or 16-bit medical images, the table becomes larger, but the idea is unchanged.

When implementing global histogram equalisation manually, one common issue is handling empty dark bins at the start of the CDF. Some implementations use a corrected formula that subtracts the first non-zero CDF value: $s_k = \text{round}(((\text{CDF}(k) - \text{CDF_min}) / (1 - \text{CDF_min})) * (L - 1))$. This prevents the darkest existing pixel from being mapped to a value above zero simply because the image has no pixels in the lowest bins. The simpler formula is easier to derive, while the corrected formula is often better in code.

For CLAHE, the implementation is more involved because there is a separate mapping for each tile. A simplified implementation would create a 3D structure where each tile has a 256-entry mapping. During pixel processing, the program identifies the surrounding tile centres, reads the corresponding mapped values, and blends them. This is why CLAHE is slower than global HE but still practical in many applications, especially with optimized libraries or parallel hardware.

Implementation detail	Why it matters
Use a luminance channel for colour images	Avoids unnatural colour shifts caused by independent RGB remapping.
Use lookup tables	Makes per-pixel remapping fast after histogram/CDF computation.
Subtract CDF_min when needed	Prevents washed-out output when low bins are empty.
Tune tile size and clip limit together	Small tiles and high clip limit are the most likely to amplify noise.
Validate with the final task	A visually better image is not always better for recognition or segmentation.

13. Extra Worked Example with CDF_min Correction

Suppose an 8-level grayscale image has intensity levels 0 to 7, but the image only uses the middle levels. The histogram counts are [0, 0, 2, 3, 5, 0, 0, 0], with 10 total pixels. The probabilities are [0, 0, 0.2, 0.3, 0.5, 0, 0, 0]. The CDF is [0, 0, 0.2, 0.5, 1.0, 1.0, 1.0, 1.0].

Level	0	1	2	3	4	5	6	7
Count	0	0	2	3	5	0	0	0
Probability	0	0	0.20	0.30	0.50	0	0	0

CDF	0	0	0.20	0.50	1.00	1.00	1.00	1.00
Simple map $\text{round}(7 * \text{CDF})$	0	0	1	4	7	7	7	7

The first non-zero CDF is $\text{CDF}_{\min} = 0.20$. If we use the corrected mapping, level 2 becomes $\text{round}(((0.20 - 0.20) / 0.80) * 7) = 0$. Level 3 becomes $\text{round}(((0.50 - 0.20) / 0.80) * 7) = 3$. Level 4 becomes $\text{round}(((1.00 - 0.20) / 0.80) * 7) = 7$. So the used input levels 2, 3, and 4 are stretched to 0, 3, and 7. This is a clearer example of contrast expansion: the narrow original range is spread over the full output range.

This example also shows why histogram equalisation can change brightness. The output does not simply preserve the average intensity; it redistributes the available levels according to the CDF. That is useful when contrast is poor, but it may be harmful when brightness consistency is important.

14. Parameter Tuning for CLAHE

CLAHE has two parameters that students often mention but do not fully explain: tile grid size and clip limit. The tile grid size decides how many local regions are used. If the grid is 8×8 , the image is divided into 64 contextual regions. If the grid is 16×16 , the method becomes more local. More local processing can reveal small structures, but it can also create a harsher image because each region has less data for a stable histogram.

The clip limit decides how much contrast amplification is allowed in each tile. A low clip limit behaves more gently. A high clip limit behaves more like ordinary adaptive histogram equalisation, which can strongly amplify noise. In medical imaging and face recognition, the best parameter choice depends on the sensor, lighting, anatomy or face position, and the later algorithm. This is why papers often perform sensitivity analysis instead of assuming one universal value.

A practical tuning workflow is: start with default CLAHE settings, inspect the result, then reduce the clip limit if noise is too strong. If the enhancement looks blocky, increase tile size or use fewer tiles. If local details are still hidden, use smaller tiles or a slightly higher clip limit. Finally, evaluate the result using the actual downstream task, not just visual appearance.

15. Comparing Global HE and CLAHE in One Paragraph

Global HE is simple and elegant: one histogram, one CDF, one lookup table. Its weakness is that it assumes the same transformation is suitable everywhere. CLAHE is more flexible: many local histograms, clipped bins, many CDFs, and interpolation. Its weakness is parameter sensitivity and the possibility of enhancing noise. In a report, the best comparison is not simply “CLAHE is better.” A stronger answer is: CLAHE is better when contrast varies spatially, but global HE can be preferable when the image is clean and uniformly under-contrasted, because it is simpler, faster, and less likely to create local artifacts.

16. Key Papers and Source Notes

1) Olmedo, E., de la Calleja, J., Benitez, A., & Medina, M. A. (2012). Point to point processing of digital images using parallel computing. *International Journal of Computer Science Issues*, 9(3), 1-10.

Direct URL: <http://ijcsi.org/papers/IJCSI-9-3-3-1-10.pdf>

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Summary and relevance: This paper explains how point processing operations in digital images can be accelerated using parallel computing with CUDA and GPUs. The authors focus on operations such as grayscale conversion, thresholding, brightening, darkening, and contrast adjustment, where each pixel is processed independently. The paper compares GPU performance against traditional CPU and OpenCV implementations and shows that CUDA significantly reduces processing time, especially for high resolution images. The work demonstrates how image processing tasks can benefit from parallel execution because millions of pixels can be processed simultaneously. This paper is useful for understanding the computational side of point processing and real time image enhancement.

It is relevant because it provides the foundation for understanding pixel-level transformations before moving to histogram-based contrast methods.

2) Reza, A. M. (2004). Realization of the Contrast Limited Adaptive Histogram Equalization (CLAHE) for real-time image enhancement. Journal of VLSI Signal Processing Systems for Signal, Image, and Video Technology, 38, 35-44.

<https://doi.org/10.1023/B:VLSI.0000028532.53893.82>

Direct URL: <https://link.springer.com/article/10.1023/B:VLSI.0000028532.53893.82>

Summary and relevance: This paper presents a hardware oriented implementation of CLAHE for real time image enhancement systems. The author explains the mathematical foundations of CLAHE, including histogram clipping, cumulative distribution functions, local tile processing, and bilinear interpolation between neighboring regions. The paper focuses on reducing computational complexity so that CLAHE can be implemented efficiently using FPGA or VLSI hardware. The proposed system uses pipelining, parallel processing, and lookup tables to accelerate image enhancement while maintaining precision. Applications discussed include medical imaging and X ray systems, where real time enhancement is critical. This paper is highly relevant because it provides detailed CLAHE derivations, implementation formulas, and practical engineering considerations.

It is highly relevant because it explains the technical mechanism that turns CLAHE from a visual idea into an implementable algorithm.

3) Musa, P., Al Rafi, F., & Lamsani, M. (2018). A review: Contrast-Limited Adaptive Histogram Equalization (CLAHE) methods to help the application of face recognition. 2018 Third International Conference on Informatics and Computing (ICIC), 1-6.

<https://doi.org/10.1109/IAC.2018.8780492>

Direct URL: <https://doi.org/10.1109/IAC.2018.8780492>

Summary and relevance: This review paper studies the use of CLAHE for improving face recognition systems under difficult lighting conditions. The authors explain that standard histogram equalisation improves global contrast but can produce unclear results in images with uneven illumination. CLAHE improves this by operating locally on small image tiles while limiting contrast enhancement to avoid excessive noise amplification. The paper combines CLAHE with Local Binary Pattern Histogram (LBPH) methods for face recognition and demonstrates that CLAHE improves recognition quality in both dark and bright conditions. The work is important because it highlights how adaptive contrast enhancement can improve biometric applications and computer vision systems that depend on image clarity. This paper reviews CLAHE in the context of face recognition and lighting variation.

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It is relevant because it links CLAHE parameters such as tile size and clip limit to a real computer vision task where uneven illumination can reduce recognition accuracy.

4) Ahmad, M. S. Z., Aziz, N. A. A., Lim, H. S., Ghazali, A. K., & Latiff, A. A. (2025). Impact of image enhancement using Contrast-Limited Adaptive Histogram Equalization (CLAHE), anisotropic diffusion, and histogram equalization on spine X-ray segmentation with U-Net, Mask R-CNN, and transfer learning. Algorithms, 18(12), 796. <https://doi.org/10.3390/a18120796>

Direct URL: <https://www.mdpi.com/1999-4893/18/12/796>

Summary and relevance: This paper investigates how image enhancement techniques affect deep learning based spine X ray segmentation. The authors compare Histogram Equalisation (HE), CLAHE, and Anisotropic Diffusion (AD) using U Net and Mask R CNN architectures. The study finds that CLAHE and AD significantly improve U Net segmentation performance, while Mask R CNN is less sensitive to preprocessing methods. CLAHE enhances local contrast by dividing the image into tiles and limiting contrast amplification through clip limits. The paper also discusses how HE may over amplify noise and how CLAHE better preserves important image details. This research is valuable because it demonstrates modern practical applications of histogram equalisation and CLAHE in medical imaging and artificial intelligence.

It is relevant because it demonstrates an important failure-case lesson: enhancement can help some models, such as U-Net, while bringing little or no improvement to stronger instance segmentation models such as Mask R-CNN.

Point Processing, Histogram Equalisation and CLAHE. (User-provided course note PDF).